



ASSIGNMENT COVER SHEET

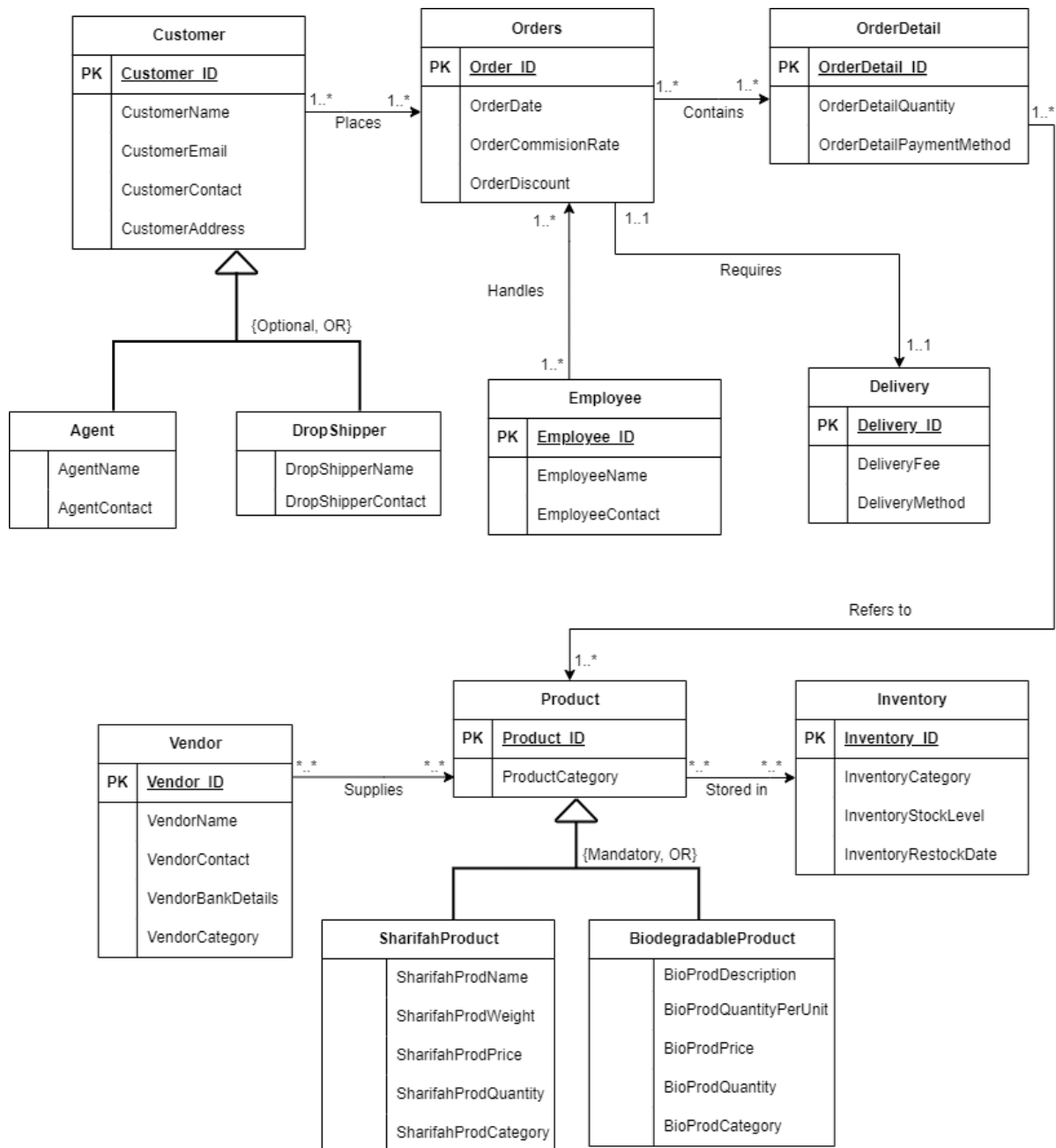
Course Code and Description	UECS1413 DATABASE SYSTEM FUNDAMENTALS		
Lecturer Name	Dr.Sugumaran a/l Nallusamy		
Assignment Title	Assignment 2		
DECLARATION			
We declare that this is a group assignment and that no part of this submission has been copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part been written for us by another person.			
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Description	Full Marks	Marks Obtained
Provide the correct or amended ERD diagram from the Programming Project Assignment I.	10	
*Remove features not compatible with the relational model *Derive relations for local logical data model.	40	
*Normalization Check The produce logical schema is expected to be in 3 rd NF. Student may normalise their tables to produce the 3rdNF version, or (if the schema in 3 rd NF, an assurance statement that the produced relational schema is in 3 rd NF is essential).	5	
List the summary of relations with attributes, primary and foreign key; A PK may be identified by an underline (a common practice) and a FK by any other means as described by the student (e.g. double underline, italic style, different colour font, etc.)	15	
Draw the logical ERD diagram Provide the logical erd diagram with entity, relationship, attributes and multiplicity. Primary key and Foreign key also presented in the diagram	10	
Define data type logically	10	
Define the size logically	5	
All the columns define in the table above are listed in the Data Dictionary	5	
Total	100	

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Original ER Diagram from Programming Project Assignment 1



Modifications to ER Diagram from Assignment 1 before proceeding to the Logical Design

We have decided to modify the entities from the ER Diagram from Assignment 1.

Firstly, we updated the participation and disjoint constraints for the ‘**Product**’ superclass and its subclass hierarchies from {Mandatory, OR} to **{Mandatory, AND}**. At the same time, we also updated the participation and disjoint constraints for the ‘**Customer**’ superclass and its subclass hierarchies from {Optional, OR} to **{Mandatory, AND}**.

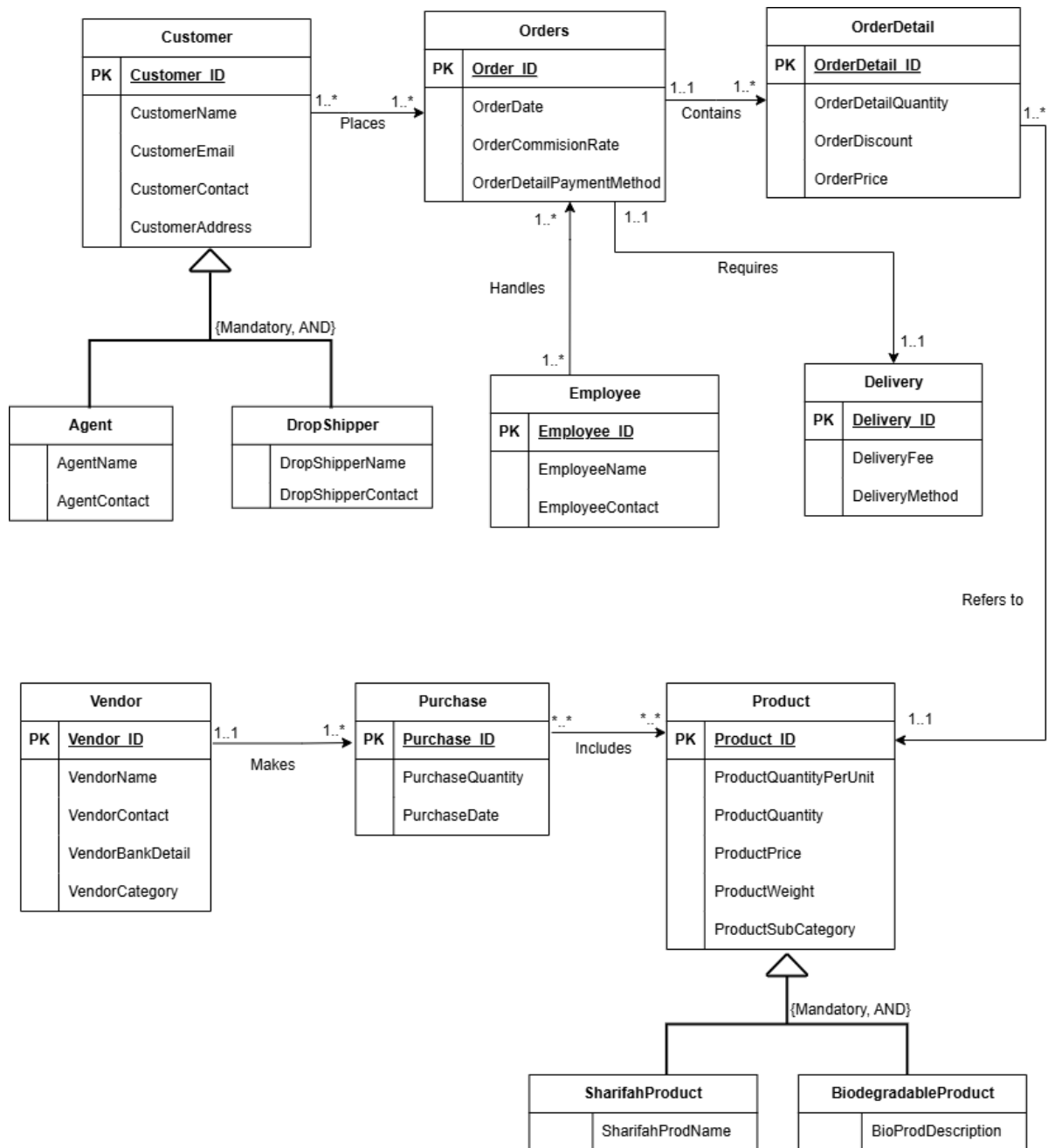
Secondly, we moved the ‘**OrderDiscount**’ attribute from the ‘**Orders**’ entity to the ‘**OrderDetail**’ entity for better organization and clarity. Similarly, the ‘**OrderDetailPaymentMethod**’ attribute was moved from the ‘**OrderDetail**’ entity to the ‘**Orders**’ entity. In the ‘**OrderDetail**’ entity, we added the ‘**OrderPrice**’ attributes to display the total price of the order placed by customers.

Thirdly, we changed the ‘**Inventory**’ entity’s name to ‘**Purchase**’ and included the attributes of **Purchase_ID** (Primary Key), **PurchaseQuantity**, and **PurchaseDate**.

Fourthly, we updated the ‘**SharifahProduct**’ entity to contain one attribute, ‘**SharifahProdName**’, and the ‘**BiodegradableProduct**’ entity to contain one attribute, ‘**BioProdDescription**’.

Lastly, we updated the ‘**Product**’ entity to include the attributes of **Product_ID** (Primary Key), **ProductQuantityPerUnit**, **ProductQuantity**, **ProductPrice**, **ProductWeight**, and **ProductSubCategory**.

Amended ER Diagram before Logical Design Process



Logical Design Process

Step 1: Remove features not compatible with the relational model

1. Remove *: * binary relationship types

There is one *: * relationship type between 'Purchase' and 'Product' entities. To remove this relationship type, we have created a new entity called 'PurchaseDetail'.

2. Remove *: * recursive relationship types

Since the ER Diagram does not contain any *: * recursive relationship types, no modifications are needed.

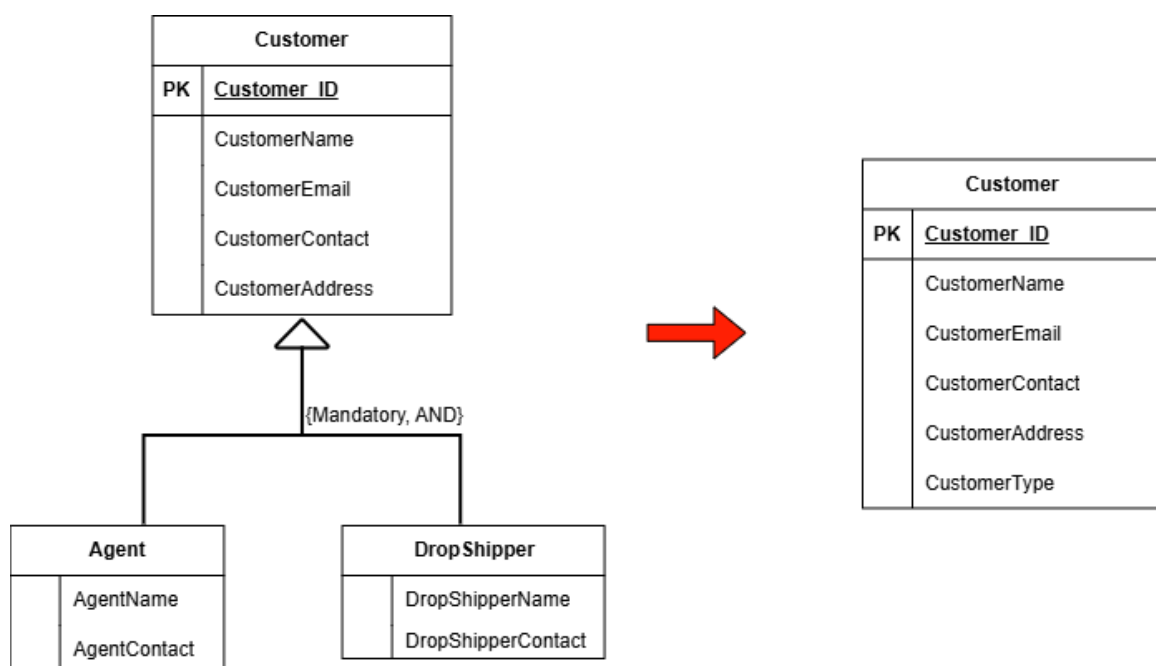
3. Remove complex relationship types

Since the ER Diagram does not contain any complex relationship types, no modifications are needed.

4. Remove superclass/ subclass relationship types

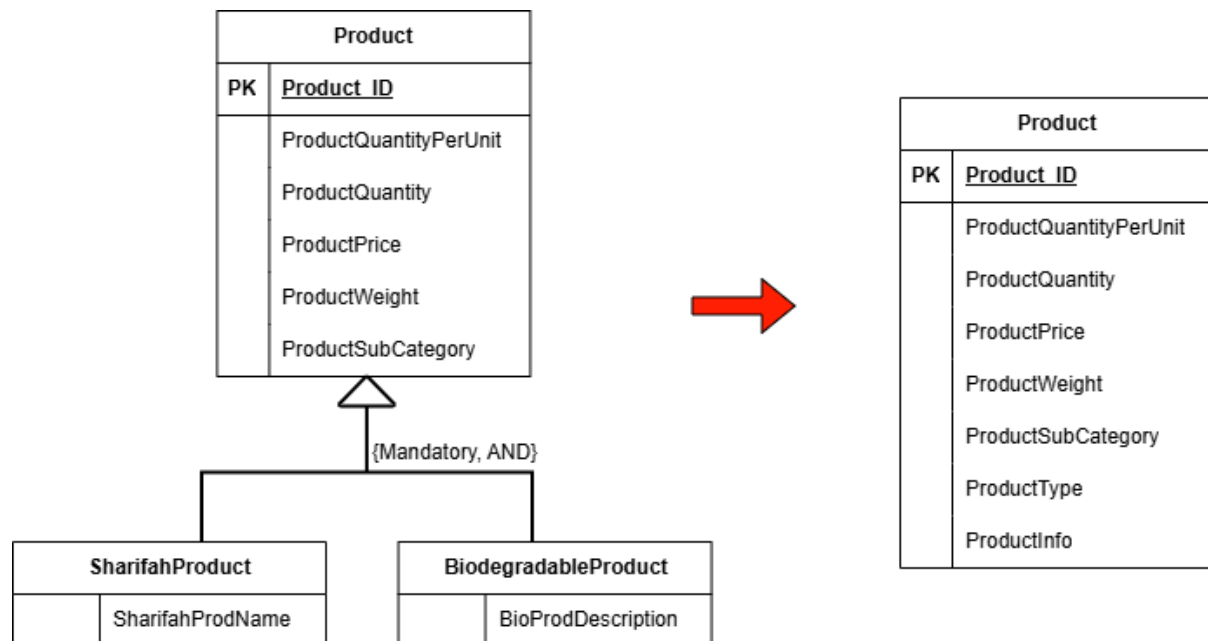
Since the 'Customer' entity contains a superclass with two subclass relationship types, we combined both 'Agent' and 'DropShipper' entities into a single relation called 'Customer'.

The 'Customer' entity includes the attributes of Customer_ID (Primary Key), CustomerName, CustomerEmail, CustomerContact, CustomerType, and CustomerAddress.



Since the 'Product' entity contains a superclass with two subclass relationship types, we combined both 'SharifahProduct' and 'BiodegradableProduct' entities into a single relation called 'Product'.

The 'Product' entity includes the attributes of Product_ID (Primary Key), ProductQuantityPerUnit, ProductQuantity, ProductPrice, ProductWeight, ProductSubCategory, ProductType.

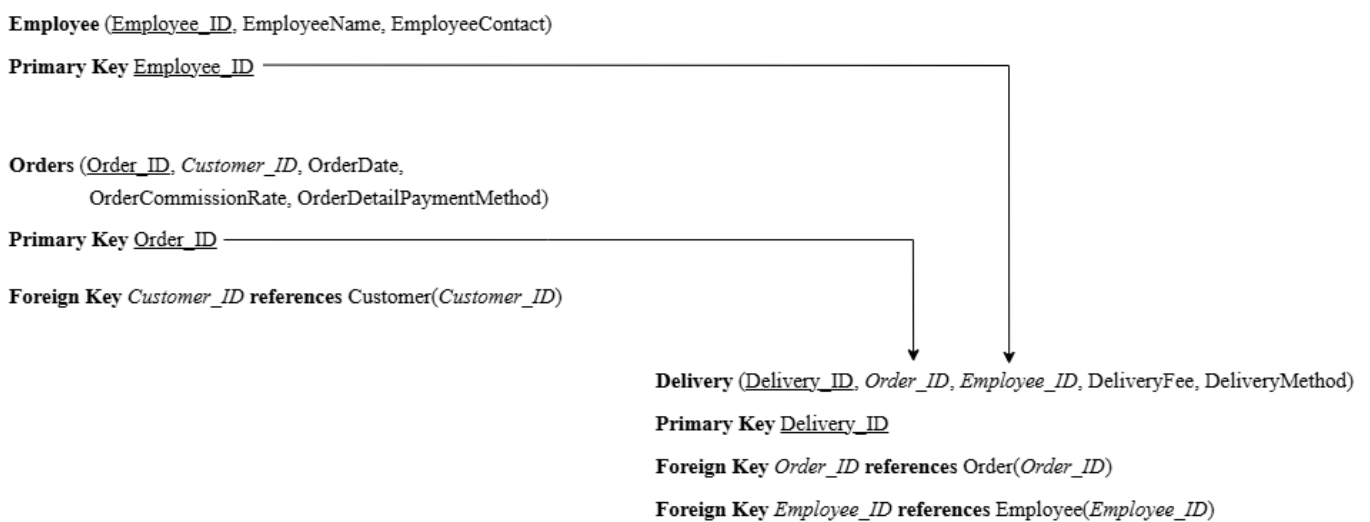
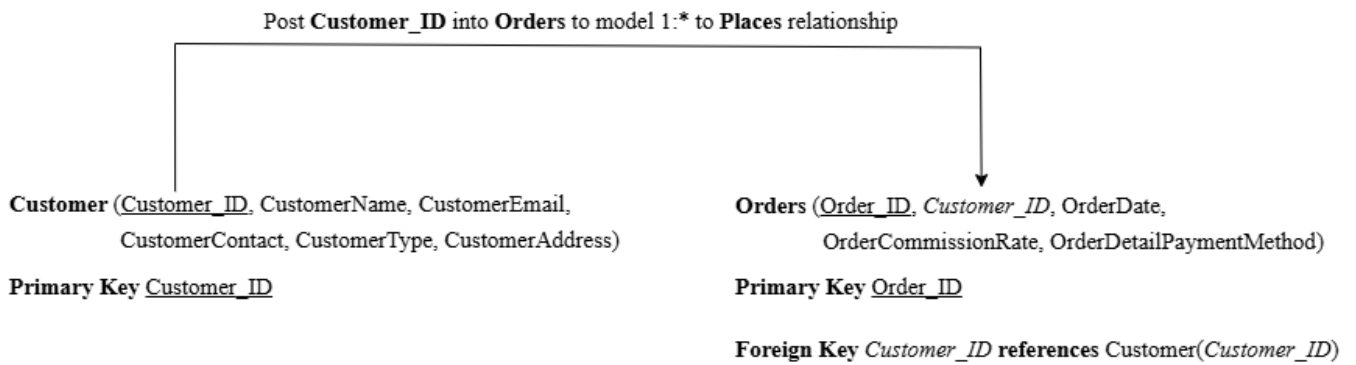
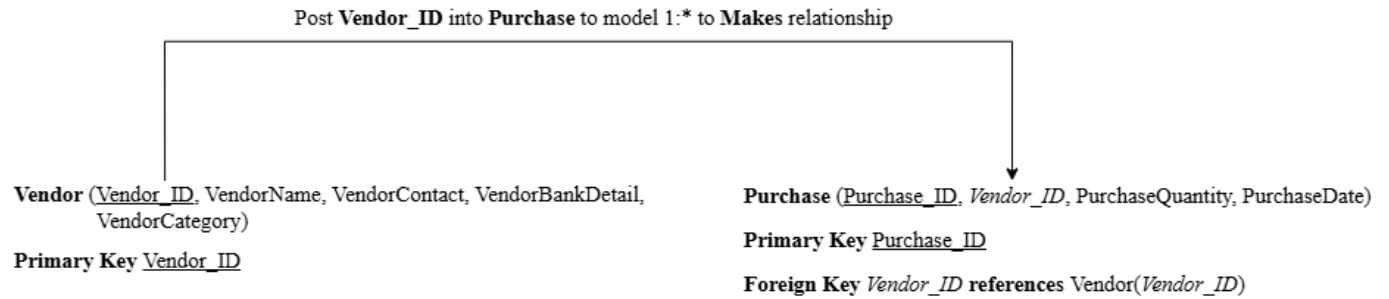


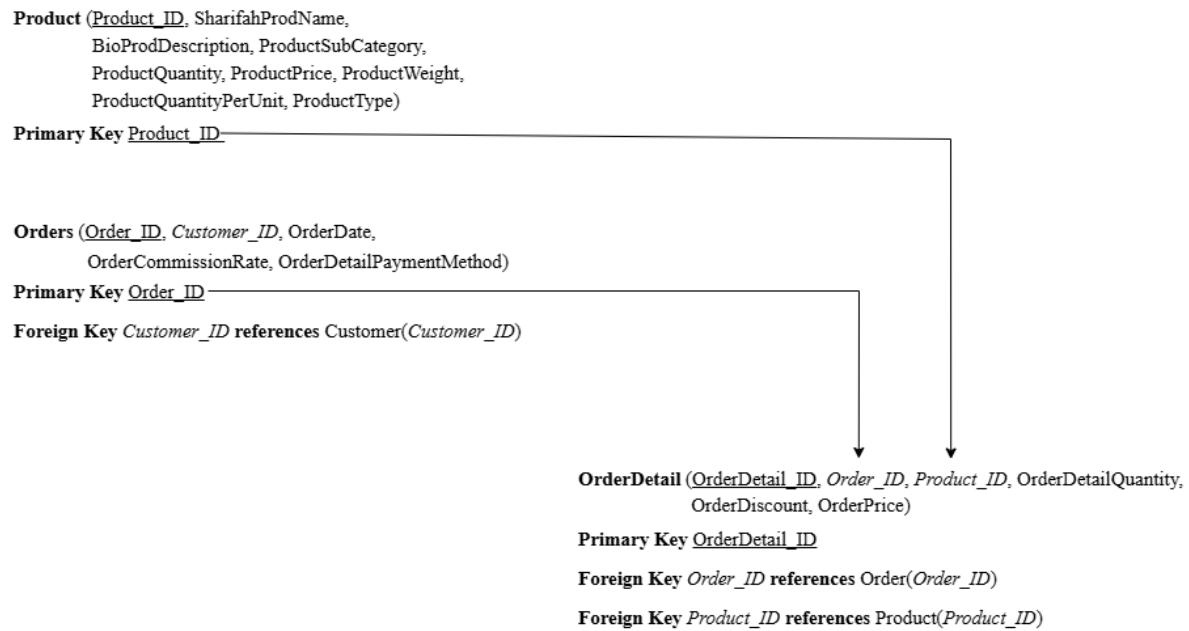
5. Remove multi-valued attributes

Since the ER Diagram does not contain any multi-valued attribute, no modifications are needed.

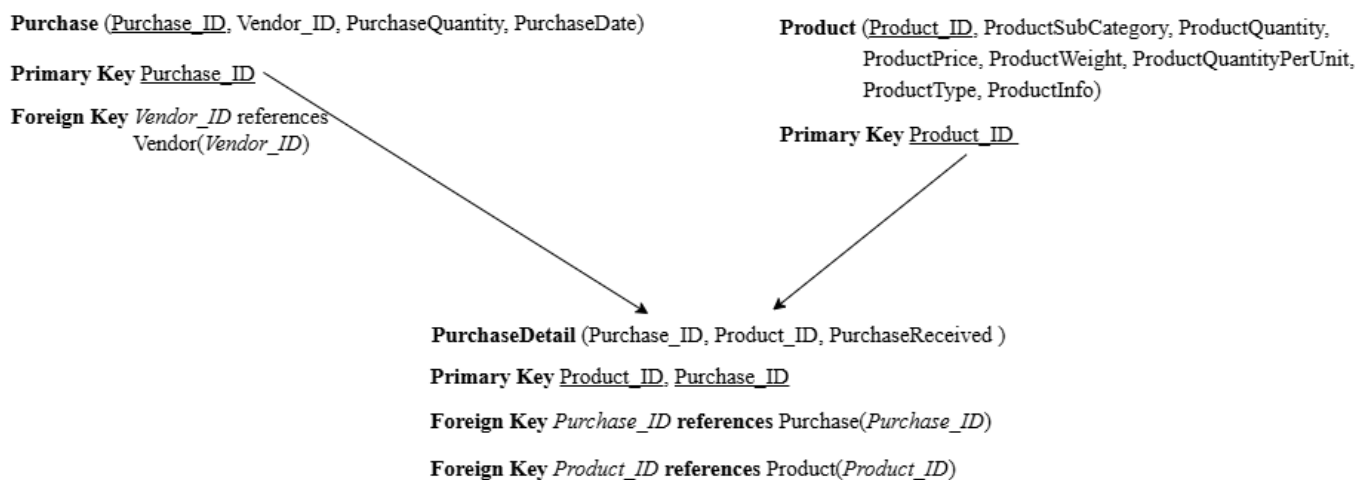
Step 2: Derive Relations for Local Logical Data Model

1. 1:* binary relationship types





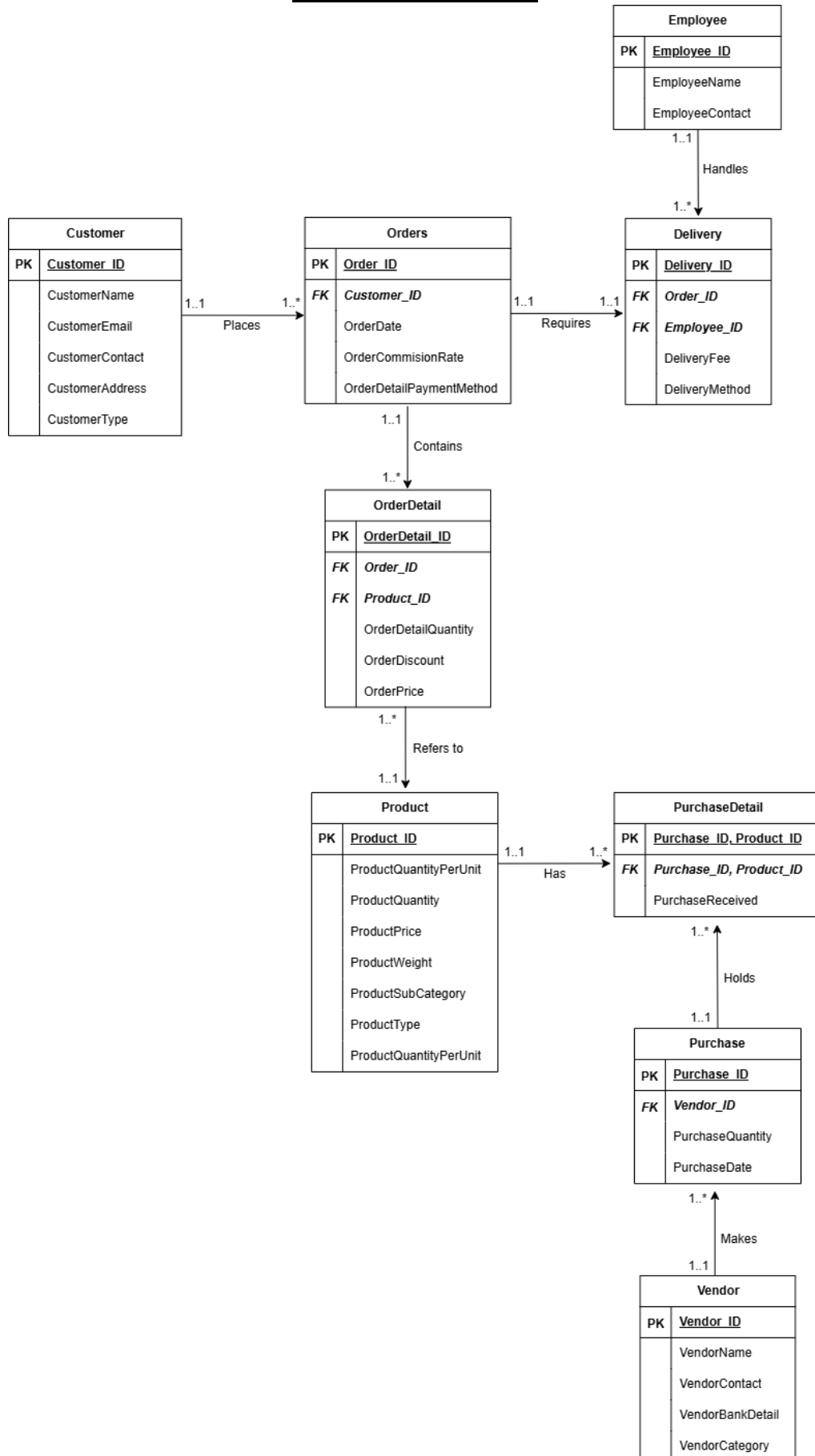
2. *: * binary relationship types



Step 3: Normalization Process

1. First Normal Form (1NF)
2. Second Normal Form (2NF)
3. Third Normal Form (3NF)

Logical ERD Diagram



Declaration

In this report, we have decided to modify some entity attributes by using short terms or names to represent the data, so that the table is easier to understand. Attributes that are not listed below will remain unchanged.

<u>Customer</u> Customer_ID → Cust_ID CustomerName → Cust_Name CustomerEmail → Cust_Email CustomerContact → Cust_Contact CustomerAddress → Cust_Add CustomerType → Cust_Type	<u>Orders</u> Order_ID → Ord_ID OrderDate → Ord_Date OrderCommissionRate → Ord_CommRate OrderDetailPaymentMethod → OD_PayMethod
<u>Employee</u> Employee_ID → Emp_ID EmployeeName → Emp_Name EmployeeContact → Emp_Contact	<u>Delivery</u> Delivery_ID → Del_ID DeliveryFee → Del_Fee DeliveryMethod → Del_Method
<u>Vendor</u> Vendor_ID → Vend_ID VendorName → Vend_Name VendorContact → Vend_Contact VendorBankDetail → Vend_BD VendorCategory → Vend_Category	<u>Product</u> Product_ID → Prod_ID ProductSubCategory → Prod_SubCategory ProductQuantity → Prod_Qty ProductPrice → Prod_Price ProductWeight → Prod_Weight ProductQuantityPerUnit → Prod_QtyPerUnit ProductType → Prod_Type ProductInfo → Prod_Info
<u>OrderDetail</u> OrderDetail_ID → OD_ID OrderDetailQuantity → OD_Qty OrderDiscount → Ord_Disc OrderPrice → Ord_Price	<u>Purchase</u> Purchase_ID → P_ID PurchaseQuantity → P_Qty PurchaseDate → P_Date
<u>PurchaseDetail</u> PurchaseReceived → P_Rec	

Summary of Relations for the System Views

Customer (<u>Cust_ID</u> , Cust_Name, Cust_Email, Cust_Contact, Cust_Type, Cust_Add) Primary Key <u>Cust_ID</u>	Employee (<u>Emp_ID</u> , Emp_Name, Emp_Contact) Primary Key <u>Emp_ID</u>
Orders (<u>Ord_ID</u> , Cust_ID, Ord_Date, Ord_CommRate, OD_PayMethod) Primary Key <u>Ord_ID</u> Foreign Key <i>Cust_ID</i> references Customer(<i>Cust_ID</i>)	Delivery (<u>Del_ID</u> , <i>Ord_ID</i> , Emp_ID, Del_Fee, Del_Method) Primary Key <u>Del_ID</u> Foreign Key <i>Ord_ID</i> references Order(<i>Ord_ID</i>)
OrderDetail (<u>OD_ID</u> , <i>Ord_ID</i> , <i>Prod_ID</i> , OD_Qty, Ord_Disc, Ord_Price) Primary Key <u>OD_ID</u> Foreign Key <i>Ord_ID</i> references Order(<i>Ord_ID</i>) Foreign Key <i>Prod_ID</i> references Product(<i>Prod_ID</i>)	Product (<u>Prod_ID</u> , Prod_SubCategory, Prod_Qty, Prod_Price, Prod_Weight, Prod_QtyPerUnit, Prod_Type, Prod_Info) Primary Key <u>Prod_ID</u>
PurchaseDetail (<u>P_ID</u> , <i>Prod_ID</i> , P_Rec) Primary Key <u>P_ID</u> , <u>Prod_ID</u> Foreign Key <i>P_ID</i> references Purchase(<i>P_ID</i>) Foreign Key <i>Prod_ID</i> references Product(<i>Prod_ID</i>)	Purchase (<u>P_ID</u> , <i>Vend_ID</i> , P_Qty, P_Date) Primary Key <u>P_ID</u> Foreign Key <i>Vend_ID</i> references Vendor(<i>Vend_ID</i>)
Vendor (<u>Vend_ID</u> , Vend_Name, Vend_Contact, Vend_BD, Vend_Category) Primary Key <u>Vend_ID</u>	

Data Dictionary

Table name: Customer

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK referenced table
Cust_ID	Unique identification for all customers	Varchar2	5	Yes	No	-
Cust_Name	Customer name	Varchar2	30	No	No	-
Cust_Email	Customer's email address	Varchar2	20	No	No	-
Cust_Contact	Customer's contact number	Number	15	No	No	-
Cust_Type	Customer's type (customer, agent, or dropshipper)	Varchar2	20	No	No	-
Cust_Add	Customer's address	Varchar2	50	No	No	-

Table name: Employee

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK referenced table
Emp_ID	Unique identification for all employees	Varchar2	5	Yes	No	-
Emp_Name	Employee name	Varchar2	30	No	No	-
Emp_Contact	Employee's contact number	Number	15	No	No	-

Table name: Orders

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK referenced table
Ord_ID	Unique identification for all orders	Varchar2	5	Yes	No	-
Ord_Date	Date of the order	Date	-	No	No	-
Ord_CommRate	Commission rate for the order	Number	5	No	No	-
OD_PayMethod	Payment method for the order	Varchar2	20	No	No	-
Cust_ID	Unique identification for all customers	Varchar2	5	No	Yes	Customer

Table name: Delivery

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK referenced table
Del_ID	Unique identification for all deliveries	Varchar2	10	Yes	No	-
Del_Fee	Delivery fees (RM)	Number	5, 2	No	No	-
Del_Method	Different types of delivery methods	Varchar2	15	No	No	-
Ord_ID	Unique identification for all orders	Varchar2	5	No	Yes	Orders
Emp_ID	Unique identification for all employees	Varchar2	5	No	Yes	Employee

Table name: OrderDetail

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK referenced table
OD_ID	Unique identification for all order details	Varchar2	10	Yes	No	-
OD_Qty	The quantity of the orders	Number	5	No	No	-
O_Disc	The discount on the orders	Number	5	No	No	-
Ord_Price	The total price of the order placed (RM)	Number	7, 2	No	No	-
Ord_ID	Unique identification for all orders	Varchar2	5	No	Yes	Orders
Prod_ID	Unique identification for all products	Varchar2	15	No	Yes	Product

Table name: PurchaseDetail

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK referenced table
P_ID	Unique identification for all purchases	Varchar2	5	Yes	Yes	Purchase
Prod_ID	Unique identification for all products	Varchar2	15	Yes	Yes	Product
P_Rec	Payments received by the Vendors (RM)	Number	7, 2	No	No	-

Table name: Product

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK referenced table
Prod_ID	Unique identification for all products	Varchar2	15	Yes	No	-
Prod_SubCategory	The product's subcategory	Varchar2	15	No	No	-
Prod_Qty	The quantity of product	Number	10	No	No	-
Prod_Price	The price of each product (RM)	Number	5	No	No	-
Prod_Weight	The weight of each product (g)	Number	5	No	No	-
Prod_QtyPerUnit	The product's quantity per unit	Number	5	No	No	-
Prod_Type	The product types (Sharifah Ready to Eat Products or Biodegradable Packaging Items)	Varchar2	30	No	No	-
Prod_Info	The product name or description for Sharifah Ready to Eat Products and Biodegradable Packaging Item	Varchar2	10	No	No	-

Table name: Purchase

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK referenced table
P_ID	Unique identification for all purchases	Varchar2	5	Yes	No	-
P_Qty	The quantity of purchase	Number	5	No	No	-
P_Date	The date of purchase	Date	-	No	No	-
Vend_ID	Unique identification for all vendors	Varchar2	5	No	Yes	Vendors

Table name: Vendor

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK referenced table
Vend_ID	Unique identification for all vendors	Varchar2	5	Yes	No	-
Vend_Name	Vendor name	Varchar2	30	No	No	-
Vend_Contact	Vendor's contact number	Number	15	No	No	-
Vend_BD	Vendor's bank account number	Number	10	No	No	-
Vend_Category	The category of the vendor (Sharifah Food Sdn Bhd or other vendors)	Varchar2	20	No	No	-